

Lab-2 Worksheet

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1. Introduction

The objective of this experiment is to design, train, and analyze a Convolutional Neural Network (CNN) using the CIFAR-10 dataset. The experiment focuses on model training, computational complexity analysis, gradient and weight flow visualization, and performance evaluation using Weights & Biases (WandB).

2. Dataset Description

The CIFAR-10 dataset consists of 60,000 RGB images of size 32×32 , distributed across 10 classes. The dataset is split into 50,000 training images and 10,000 testing images. Basic normalization was applied for stable training.

3. Custom DataLoader

A custom PyTorch Dataset class was implemented to load CIFAR-10 images. The DataLoader performs:

- Conversion to tensor
- Normalization using mean and standard deviation
- Batch loading with shuffling for training

4. CNN Model Architecture

A SimpleCNN architecture was used consisting of:

- Two convolutional layers with 32 and 64 filters
- ReLU activation functions
- MaxPooling layers
- Two fully connected layers for classification

5. FLOPs and Parameter Analysis

The computational complexity of the model was calculated using the ptflops library.

Total Parameters: 1.07 Million

FLOPs: 6.9 Million MACs

Most parameters lie in fully connected layers, while most computation occurs in convolution layers.

6. Training Configuration

The model was trained using:

Optimizer: Adam

Learning Rate: 0.001

Batch Size: 64

Epochs: 30

Loss Function: Cross Entropy Loss

7. Gradient Flow and Weight Update Visualization

Gradient flow visualizations showed stable gradients across layers with no vanishing or exploding gradient issues.

Weight update flow plots showed smooth and consistent parameter updates across epochs.

8. Results

The model achieved steady improvement during training.

Final Training Accuracy: 99.12%

Final Test Accuracy: 71.64%

9. Observations and Analysis

- The model learned effectively as seen from high training accuracy.
- The test accuracy is lower, indicating slight overfitting.
- Gradient flow remained stable throughout training.
- Weight update flow indicated smooth learning behavior.

10. Overfitting Discussion

The gap between training and test accuracy suggests that the model memorized training data.

This can be improved using regularization or data augmentation.

11. WandB Reporting

All metrics such as training loss, accuracy, gradient flow, weight flow, and FLOPs were logged to WandB.

WandB dashboards enabled real-time monitoring of experiments.

12. Links

Github Link: https://github.com/mamta54/MLOps-Mamta-M25CSA018/tree/lab2_worksheet

WandB Link: <https://wandb.ai/mamta000542-prom-iit-rajasthan/cifar10-cnn-mlops>

Conclusion

This experiment demonstrates a complete MLOps workflow including CNN training, FLOPs analysis, gradient monitoring, and experiment tracking using Weights & Biases.